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| **Project Title:** | Analysis of the Gravisensing System in Chara | | |
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| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | |
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| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA ARC |
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This proposal focuses on the study of plant gravisensing in the Chara rhizoid through magnetophoretic manipulation of statoliths. Despite the long-standing correlation between the sedimentation of dense organelles or even the entire protoplast and subsequent growth responses, little is known about the transduction of the stimulus or the mechanism by which the position of the putative sensors is maintained.
Based on a previously developed magnetic system that provides controllable ponderomotive forces specific for the statoliths, we will analyze the factors that determine the sedimentation of statoliths and the viscoelastic friction and anchorage of the cell interior. Although earlier work has indicated that F-actin may be instrumental for the spatial distribution of the Chara statoliths, we don't know whether the position of the statoliths, their movement, or the force that they exert on cytoskeletal elements represents the actual signal. The model system of the Chara rhizoid is ideal for these studies as within a single cell the force action and response can be observed. These experiments will be based upon the controlled displacement of the statoliths by adjustable high gradient magnetic fields (HGMF), which can exert

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| **Task Description:** | a ponderomotive force of >2 g and, depending upon the mutual position of the magnetic poles and statoliths, can act with or against gravity. Therefore, we can mimic hypo- and hypergravity. The mechano-elastic properties of cytoskeletal components will be analyzed by measuring the extent and time course of displacement of statolith clusters under different force values. These measurements will determine if directional growth is controlled by pressure gradients, position or movement (sedimentation) of statoliths. By manipulating the degree of polymerization of F-actin (shifting the equilibrium to g-actin with cytochalasin B or D or latrunculin and to more stable F-actin with phalloidin) or microtubules (depolymerized with propizamide or oryzalin or stabilized with taxol) the effect of the dominant cytoskeletal components on the viscoelastic property of the cytoplasm and positioning of the statoliths will be determined.<br><br>The results of the proposed experiments will deepen our understanding of gravity sensing in plants and the role that the cytoplasm plays in the mechanics of perception and signal transduction. The investigations will identify whether a hitherto neglected aspect, namely changes in the viscoelastic properties of the cytoskeleton, constitutes an integral part of the gravisensing mechanism. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The work will analyze in vivo the mechanical properties of the cytoplasm and cytoskeleton. In the context of gravisensing and response of the Chara rhizoids. High gradient magnetic fields (HGMF)can be used to exert a ponderomotive force directly on the statoliths and the analysis of the time course of the resulting displacement will allow a determination of the elastic modulus and visosity of the cytoplasm. By selectively disrupting the components of the cytoskeleton, the measurements will reveal the relative participation of microfilaments and/or microtubules in the transduction process. The most noticable benefits of the research include a better understanding of plant growth regulation and signal transduction events. These processes can help improve growth, resilience, and response to environmental changes of (crop) plants. |
| **Task Progress:** | During gravistimulation, several events alter growth of the single cell rhizoids of the green algae Chara. The dense statoliths in the rhizoids tip sediment when the rhizoid is oriented in a horizontal position. When the statoliths make contact with the plasma membrane, a signal is induced that leads to the alteration of the rhizoids growth. The rhizoid curves until is grows again parallel to the gravity force.<br>Our experiments rely on the manipulation of rhizoids by microinjection of magnetic particles of the size of the endogenous statoliths that can be manipulated with a magnet from outside the cell. We successfully established this method and are now able to injection different solutions and particles into the single cell rhizoid.<br><br>Chara cultures<br><br>The rhizoids are derived from freshly germinate Chara oocytes that best grow in a combination of pond soil and tap water with ca. 15 - 200 C, artificial light in an 16h : 8h day : night regime).<br><br>2. Hardware-Setup<br><br>Microscope<br><br>The best method was developed by fixing the rhizoid with a patch pipette (obtained from a special program on a Sutter P-2000 puller) and a second micromanipulator to the microscope (modified Nikon Eclipse E 600 FN). Experiments are typically performed under a long-distance objective (50X). A second horizontal microscope was custom-built from Infini-tube, eye-piece holder and equipped with a Mitutoyo long working distance objective (80x, working distance = 15 mm). The stage holds a micro cuvette (e.g., 3 x 3 x 45 mm) and is mounted to a 3D stage holder. The stage can be rotated 360° to reorient the rhizoid. Microinjection-Pipettes<br><br>The pipette is one of the most critical pieces in the microinjection and is a compromise between minimizing penetration-induced damage to the rhizoid and passage of the beads through the tip. The best results were obtained with thin-walled borosilicate glass and 5 - 7 µm inner and 7 - 9 µm outer diameter. The pipettes were beveled on a commercial glass grinder with lapping film (0.1 µm particle size). Beveled tips are silanized in dichloro-dimethyl-silane at 60 °C for 24 h. Pipettes are backfilled with the bead suspension in injection buffer.<br><br>Magnetic beads<br><br>To compare the effect of endogenous statoliths and injected beads, we choose beads of 1 ?m diameter with ferrous core and latex coating. We developed a method to keep beads separated during injection and are now able to inject as little as 3 - 5 beads.<br><br>Results<br><br>Induction of curvature using magnetic beads<br><br>The rhizoid grew in a vertical orientation in the presence of a magnet that attracted the injected magnetic particles. The magnetic beads moved toward the magnet and the rhizoid transiently grew toward the magnet. After 48 h the rhizoid grew parallel to the gravity vector. The adjustment was dependent on the number of injected beads. When more than about 15 beads were injected the rhizoid grew straight toward the magnet, fewer beads resulted in a transient growth path. A research paper is in preparation which reports the above mentioned results.<br><br>In vivo measurements of cytoplasmic viscosity<br><br>Moving injected superparamagnetic beads provides a means to measured cytoplasmic viscosity in various regions of the rhizoid. The measurements were performed by dislocating the beads with a magnet at and for the same distance that was used to calibrate the system (movement of beads in glucose and sucrose solutions of known viscosities). The cytoplasmic viscosity varied by two orders of magnitude. Application of the actin-depolymerizing drug latrunculin B reduced the viscosity. These data represent the first mapping of viscosity of a living plant cell.<br><br>The lowest viscosity was cytoplasmic streaming (1.1 centiPoise, cP), similar to water (1 cP). The cytoplasm around the vacuole showed a significantly higher viscosity (43 cP). The lateral viscosity was higher in the region between nucleus and statolith-zone, and beads were more readily displaced in the longitudinal direction (viscosity = 2 cP) compared to a |

lateral direction (14 cP). The highest viscosity in this zone was found along the cell wall (30 cP). The volume around the endogenous statoliths showed longitudinal viscosity of ~100 cP and lateral viscosity of 60 cP. A research paper of these results is in preparation.

Microinjection of poly dT coated magnetic beads and extraction of beads/mRNA and the development of Solid Phase Gene Extraction

The injection of oligo dT coated magnetic beads of 1 µm diameter into the tip of rhizoids for 2h, 4h and 6h with subsequent extraction, and reversed transcription into cDNA was used to study the mRNA microextraction.

Compared to 2h incubation, 4h reduced detectable mRNA to half and only traces were detectable after 6h. This study prompted us to develop a novel mRNA extraction method we named Solid Phase Gene Extraction. This method uses oligo dT nucleotides that are covalently bound to the tip (0.5 µm diameter) of a borosilicate glass needle. It allows extracting mRNA from tissue or sufficiently large single cells without destroying/killing the sampled tissue or cell. Therefore, mRNA can be extracted from the same tissue/cell with a temporal and special resolution. A patent for Solid Phase Gene Extraction is pending. SPGE represents a novel method for mRNA extraction from a single cells or tissue that is non-destructive and therefore allows multiple sampling. The sensitivity of SPGE exceeds other extraction procedures in quantity (10-16 g/°l) and quality and has enabled an until now not reported temporal and spatial analysis of gene expression in a single cell. Most importantly, the difference in gene expression at different locales in the same cell illustrates that even vigorous cytoplasmic streaming does completely randomize mRNA. These results are being prepared for publication.

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